EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **email**.
- The form will use both **GET** and **POST** methods for data submission.

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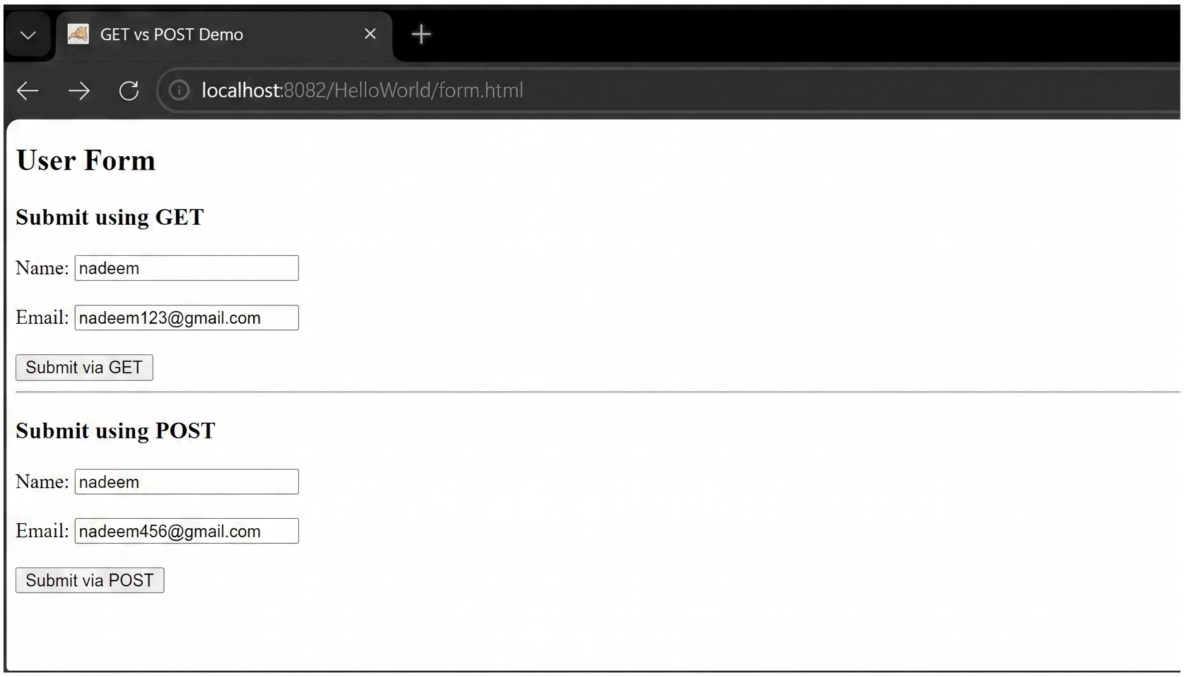
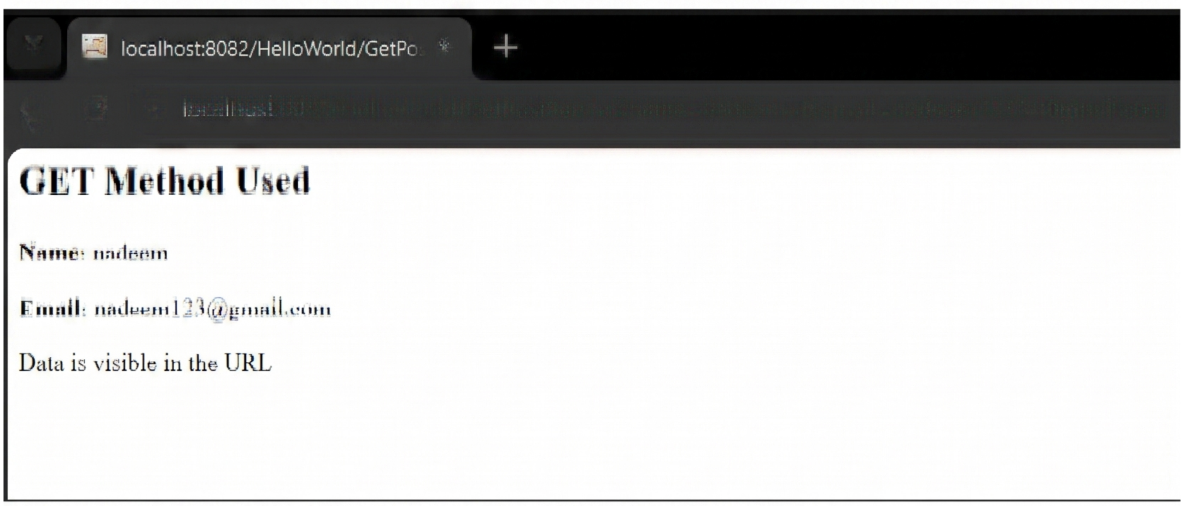
2. Steps for Creating and Handling the HTTP GET and POST Requests:

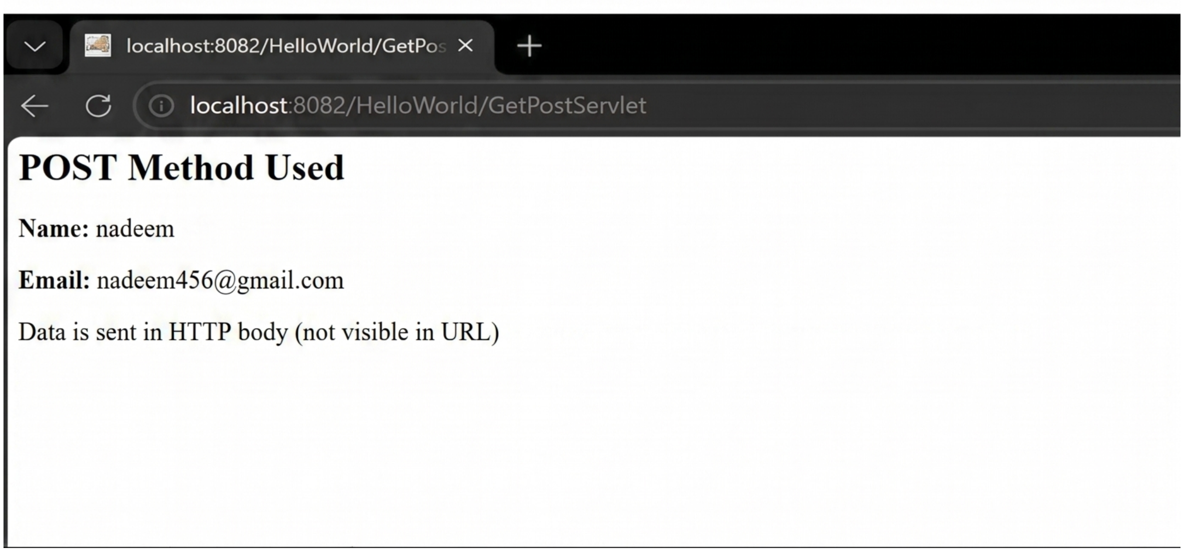
1. **Create an HTML Form:**
   - Design an HTML form with two input fields: one for the **name** and one for the **email**.
   - Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
   - Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).
2. **Create the Servlet to Handle GET and POST Requests:**
   - Write a Java Servlet class that will handle **both GET and POST requests**.
   - The servlet will have two methods:
     - doGet() to handle the GET requests.
     - doPost() to handle the POST requests.
   - In both methods, retrieve the form data using request.getParameter().
   - In the doGet() method, process and display the data in the URL.
   - In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
    - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
    - Output the data as a response, including the **name** and **email** in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
    - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
    - Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).
5. **Return the Output:**
    - The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
    - For the **GET method**, the output should be visible in the URL.
    - For the **POST method**, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
    - Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Test the form using **GET** and **POST** methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**
    - Test the **GET method**: The form data is sent in the URL.
    - Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

Output:


GET vs POST Demo
localhost:8082/HelloWorld/form.html
User Form
Submit using GET
Name: nadeem
Email: nadeem123@gmail.com
Submit via GET
Submit using POST
Name: nadeem
Email: nadeem456@gmail.com
Submit via POST


localhost:8082/HelloWorld/GetPo
GET Method Used
Name: nadeem
Email: nadeem123@gmail.com
Data is visible in the URL


localhost:8082/HelloWorld/GetPos
localhost:8082/HelloWorld/GetPostServlet
POST Method Used
Name: nadeem
Email: nadeem456@gmail.com
Data is sent in HTTP body (not visible in URL)

```html
<!DOCTYPE html>
<html>
<head>
   <title>GET vs POST Demo</title>
</head>
<body>
   <h2>User Form</h2>

   <!-- GET Form -->
   <form action="GetPostServlet" method="get">
      <h3>Submit using GET</h3>
      Name: <input type="text" name="name"><br><br>
      Email: <input type="email" name="email"><br><br>
      <input type="submit" value="Submit via GET">
   </form>

   <hr>

   <!-- POST Form -->
   <form action="GetPostServlet" method="post">
      <h3>Submit using POST</h3>
      Name: <input type="text" name="name"><br><br>
      Email: <input type="email" name="email"><br><br>
      <input type="submit" value="Submit via POST">
   </form>

</body>
```

</html>

Java File (GetPostServlet.java)

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class GetPostServlet extends HttpServlet {

  // Handle GET request
  protected void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String name = request.getParameter("name");
    String email = request.getParameter("email");

    out.println("<html><body>");
    out.println("<h2>GET Method Used</h2>");
    out.println("<p><b>Name:</b> " + name + "</p>");
    out.println("<p><b>Email:</b> " + email + "</p>");
    out.println("<p>Data is visible in the URL</p>");
    out.println("</body></html>");
  }
```

```
    // Handle POST request
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");
        out.println("<h2>POST Method Used</h2>");
        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Email:</b> " + email + "</p>");
        out.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
        out.println("</body></html>");
    }
}
```

File (web.xml):

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
         https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
         version="6.0">
```

```
    <servlet>
        <servlet-name>GetPostServlet</servlet-name>
        <servlet-class>GetPostServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>GetPostServlet</servlet-name>
        <url-pattern>/GetPostServlet</url-pattern>
    </servlet-mapping>
</web-app>
```

Java Compiled File (GetPostServlet.class) :

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

public class GetPostServlet extends HttpServlet {
   public GetPostServlet() {
   }

   protected void doGet(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {
      var2.setContentType("text/html");
      PrintWriter var3 = var2.getWriter();
```

```
        String var4 = var1.getParameter("name");

        String var5 = var1.getParameter("email");

        var3.println("<html><body>");

        var3.println("<h2>GET Method Used</h2>");

        var3.println("<p><b>Name:</b> " + var4 + "</p>");

        var3.println("<p><b>Email:</b> " + var5 + "</p>");

        var3.println("<p>Data is visible in the URL</p>");

        var3.println("</body></html>");

    }


    protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {

        var2.setContentType("text/html");

        PrintWriter var3 = var2.getWriter();

        String var4 = var1.getParameter("name");

        String var5 = var1.getParameter("email");

        var3.println("<html><body>");

        var3.println("<h2>POST Method Used</h2>");

        var3.println("<p><b>Name:</b> " + var4 + "</p>");

        var3.println("<p><b>Email:</b> " + var5 + "</p>");

        var3.println("<p>Data is sent in HTTP body (not visible in URL)</p>");

        var3.println("</body></html>");

    }

}
```

RESULT:

Thus, A Servlet was created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.